



ROOT CAUSE ANALYSIS

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

SDDC Manager Backup Recovery Stale Inventory MoRefs

The file-based backup fails after an out-of-band vCenter/NSX redeploy — a wiped NSX service account and systemic stale Managed Object References. Full diagnosis, the fixes that landed, and the correct handoff to the supported reconcile tool.

v1.1

VCF 9.0.1 — Resolved

© 2026 Virtual Control LLC — All Rights Reserved

PROPRIETARY & CONFIDENTIAL

SDDC Manager Backup Recovery — Stale Inventory MoRefs

Prepared by: Virtual Control LLC **Date:** July 10, 2026 **Document Version:** 1.1 — **RESOLVED** **Product baseline:** VMware Cloud Foundation 9.0.1 (SDDC Manager 9.0.1.0, PostgreSQL 15) **Classification:** Internal

Status: RESOLVED 2026-07-10 — the SDDC Manager backup completes end-to-end (all 8 sub-tasks SUCCESSFUL, written to SFTP). Section 6 is the full, reproducible recovery procedure.

One-paragraph summary. After a vCenter (and later NSX) instance was **redeployed outside SDDC Manager**, the SDDC Manager file-based backup failed. This is not one bug — it is **layered inventory damage**. Layer 1: the SDDC→NSX **service account was wiped** from NSX, so `operationsmanager 403s` and `/v1/system/health` returns 500. Layer 2 (the deeper one): the redeploy **re-minted every vCenter Managed Object Reference (MoRef)**, and SDDC Manager's inventory still held the **old, now-invalid MoRefs**. Because SDDC derives display names by resolving those MoRefs against live vCenter, a stale MoRef yields **no name**, and the backup's metadata exporter crashed with an unguarded `KeyError`. **Three verified fixes cleared it** (NSX account recreate; datastore MoRef `25→16` ; four VDS port-group MoRefs `56-59→45-49`) — all with values read live from vCenter, no guessing; the entitled reconcile tool was not required. A residual health/backup-status **500** remains from other stale refs (§6.7) but does not affect the backup.

1. Symptom and Impact

- **Observed:** On-demand and scheduled SDDC Manager backups fail. `POST /v1/backups/tasks` → task ends `SDDC_MANAGER_BACKUP_FAILED`.
- **Secondary symptoms:** `GET /v1/system/health` → **HTTP 500**; `GET /v1/nsxt-clusters` shows `vip: null`; `GET /v1/clusters` shows `primaryDatastoreName: null`.
- **Impact:** No valid SDDC Manager backup can be produced. This blocks a supported restore point for the management domain. Workloads and NSX data plane are **unaffected and stable** — this is a management/lifecycle-plane condition only.
- **Trigger:** The management-domain **vCenter was redeployed out-of-band** (Host-Client OVA path) during earlier recovery, and the **NSX Manager was later redeployed** during the transport-node rebuild. Neither redeploy went through SDDC Manager, so SDDC's inventory was never reconciled to the new object identities.

2. Root-Cause Architecture

The backup fails for two independent reasons, in two layers. Fixing the first exposes the second.

Layer 1 — Wiped NSX service account (authentication). The account SDDC Manager uses to log in to NSX — `svc-sddc-manager-nsx-vip-9971` — was **deleted from NSX** by the out-of-band NSX redeploy. operationsmanager therefore receives **403 Unauthorized** on every NSX call, which surfaces as health 500, a null NSX VIP, and a failure of the backup's NSX-dependent configuration step.

Layer 2 — Systemic stale MoRefs (data consistency). A vCenter redeploy **re-mints every Managed Object Reference** (`datastore-NN` , `dvportgroup-NN` , `host-NN` , ...). SDDC Manager's inventory still stores the **pre-redeploy** MoRefs. SDDC **does not store most display names — it derives them at read time by resolving the stored MoRef against live vCenter**. A stale MoRef resolves to nothing, so the derived name is absent, and the backup's metadata exporter reads it with an unguarded dictionary access.

The precise failure mechanism (Layer 2). The backup module `/opt/vmware/sddc-support/backup/sddcmanagerbackup.py` , function `export_metadata` , builds `metadata.json` and reads fields such as `cluster['primaryDatastoreName']` and `port_group['name']` **without a null/ KeyError guard**. When the underlying MoRef is stale, that key is missing and the function raises `KeyError` , which fails the sub-task `BackupSDDCManagerSystemConfiguration` → the whole backup fails. Broadcom KB **429861** documents this exact `KeyError: 'primaryDatastoreName'` signature and routes it to **Support** (no self-service SQL is published).

3. Diagnostic Evidence

3.1 Backup sub-task progression

The 8 sub-tasks run in order. Tracking where the failure lands proved each fix advanced the backup:

#	Sub-task	Before any fix	After NSX-acct fix	After datastore-MoRef fix
1	QuiesceSystem	(blocked early)	SUCCESSFUL	SUCCESSFUL
2	BackupSDDCManagerDatabase	—	SUCCESSFUL	SUCCESSFUL
3	BackupSDDCManagerConfiguration	FAIL (NSX 403)	SUCCESSFUL	SUCCESSFUL
4	BackupSDDCManagerSystemConfiguration	pending	FAIL (KeyError: primaryDatastoreName)	FAIL (KeyError: 'name')
5–7	Package / Save→SFTP / DeleteOld	pending	pending	pending
8	UnquiesceSystem	—	SUCCESSFUL	SUCCESSFUL

Note the failure **moved** from a 403 (auth) to `KeyError: primaryDatastoreName` (datastore MoRef) to `KeyError: 'name'` (port-group MoRef) — three distinct causes, each unmasked by fixing the prior one. Locks return to **0** after each run (no scheduler jam once the account exists).

3.2 The SOS backup log (authoritative error)

Each backup writes its own log under `/var/log/vmware/vcf/sddc-support/backup-<timestamp>-<pid>/sos.log`. The relevant lines:

```
ERROR [vcf_sos] [sddcmanagerbackup.py::export_metadata::236] 'primaryDatastoreName'
KeyError: 'primaryDatastoreName'
...
ERROR [vcf_sos] [sddcmanagerbackup.py::export_metadata::255] 'name'
KeyError: 'name'
Exception: Exporting SDDC Manager metadata operation failed
errorCode: BACKUP_SDDC_MGR_SYSTEM_CONFIG_FAILED
```

3.3 Confirming names are MoRef-derived (the key insight)

Disassembling `export_metadata` (source ships as `.pyc`) showed line 255 is **not** about hosts or datastores — it iterates a VDS's port groups and reads the management port group's name:

```
for port_group in switch['portGroups']:
    if port_group['transportType'] == VM_MANAGEMENT:
        data[STR_PORT_GROUP] = port_group['name']    # KeyError: 'name'
```

The internal inventory (`GET http://localhost/inventory/vds?id=...`) returns those port groups **with the MoRef (sourceId) present but no name key** — and the MoRefs are stale (`dvportgroup-56-59` , absent from current vCenter).

Proof of the derive-from-MoRef model: after the cluster's datastore MoRef was corrected (§4.2), `primaryDatastoreName` *started resolving* (`null` → `vcenter-cl01-ds-vsan01`). Writing a `name` into the inventory via `PUT /inventory/vds/{id}` returns 200 but the service **strips it** — confirming `name` is derived, not stored.

4. Fixes Applied (verified values only)

4.1 NSX service account — recreated

Covered in full by the companion runbook “**Recreating a Missing SDDC Manager → NSX Service Account” (v1.0)**. In brief: the account is **password-based** (not a certificate/principal identity); the exact username + password were read from SDDC's own credential store (`GET https://127.0.0.1/v1/system/credentials/service` , KB 327195); the NSX node service user was recreated with `POST /api/v1/node/users?action=create_service_user` ; its auto-created `auditor` role binding was promoted to `enterprise_admin` via `PUT /api/v1/aaa/role-bindings/{id}` ; a basic-auth test returned **HTTP 200**; and after `systemctl restart operationsmanager` the 403 cleared and sub-task 3 began passing.

4.2 Cluster primary datastore MoRef — corrected in the platform DB

Verified value, single row. vCenter reports the vSAN datastore `vcenter-cl01-ds-vsan01` as MoRef `datastore-16` ; SDDC's `cluster` table held the pre-redeploy `datastore-25` . Using the trust-auth workaround (see the Resource-Lock DB-Cleanup runbook for the `pg_hba.conf` toggle on `/data/pgdata` , PostgreSQL 15):

```
UPDATE cluster SET primary_datastore_source_id='datastore-16'
WHERE id='dc5f4a5b-4ab2-41d9-9e7a-5e7d84ab2712';
```

Result: UPDATE 1 ; pg_hba.conf restored to scram-sha-256 ; after sddcmanager_restart_services.sh the API's primaryDatastoreName resolved correctly and the backup advanced past sub-task 4's first KeyError . This matches the mechanism in KB **444862** (primary_datastore_source_id correction).

5. Resolution

RESOLVED 2026-07-10 — the backup completes end-to-end: all 8 sub-tasks SUCCESSFUL, written to SFTP, locks 0. Three fixes cleared it, all with **values verified live from vCenter** (never inferred): (A) the wiped NSX service account was recreated; (B) the cluster's primary-datastore MoRef was corrected `datastore-25` → `datastore-16` ; (C) the four VDS port-group MoRefs were corrected `dvportgroup-56-59` → `45-49` . After a services restart, the exporter resolved the VM_MANAGEMENT port-group name and the backup ran to completion.

The entitled `vcf-async-patch-tool --sync` was NOT required — targeted manual MoRef corrections were sufficient. The VDS's own MoRef (`dvs-54`), the vCenter `datastoreForVmDeploymentSourceId` (`datastore-25`), and host MoRefs remain stale but are **not read by the backup**, so they do not block it; they can be reconciled later by the async patch tool (or ignored). The full, reproducible procedure is Section 6.

5.1 The port-group MoRefs (the final blocker)

VDS <code>dvs-54</code> port group	SDDC inventory (stale)	Corrected to (live vCenter)	Resolves to name
VM_MANAGEMENT <i>(the one the backup reads)</i>	<code>dvportgroup-59</code>	<code>dvportgroup-49</code>	<code>pg-vm-mgmt</code>
MANAGEMENT	<code>dvportgroup-56</code>	<code>dvportgroup-45</code>	<code>pg-mgmt</code>
VMOTION	<code>dvportgroup-57</code>	<code>dvportgroup-46</code>	<code>pg-vmotion</code>
VSAN	<code>dvportgroup-58</code>	<code>dvportgroup-47</code>	<code>pg-vsan</code>

Corrected via the inventory-PUT method (§6.4). **Key subtlety:** `name` is *derived* (not stored), so the inventory GET never shows it even after the fix — the exporter resolves it live from the corrected `sourceId` , so success is verified by re-running the **backup**, not by re-reading the inventory.

6. Full Recovery Procedure (reproducible — no step omitted)

Re-derive every MoRef and ID live — do NOT reuse the literal values in this runbook. A future redeploy re-mints *different* MoRefs. The **method** is reusable; the **values** (`datastore-16` , `dvportgroup-49` , the

cluster/VDS UUIDs) must be re-read from live vCenter and this environment each time. Every value shown was verified live on 2026-07-10.

6.0 Access and conventions

- **SDDC API (from a workstation)** — get a bearer token, then call `https://sddc-manager.lab.local/v1/... : TOKEN=$(curl -sk -X POST https://sddc-manager.lab.local/v1/tokens \ -H "Content-Type: application/json" \ -d '{"username":"administrator@vsphere.local","password":"<sso-pw>"}' \ | python -c "import sys,json;print(json.load(sys.stdin)['accessToken'])"`
- **SDDC shell** — `ssh vcf@sddc-manager.lab.local`, then `su -` for root. The restricted console **corrupts long/multi-line pastes** (drops the first character; injects spaces when a line wraps). Keep every pasted line short, or drive commands over `plink` from a workstation.
- **Internal inventory API** (on the appliance, `localhost`, no auth): `curl http://localhost/inventory/{clusters,vcenters,hosts,vds}` — GET-by-id uses `?id=<uuid>`; PUT uses `/<uuid>` with a single-**object** body.
- **vCenter REST (for live MoRefs)**: `SID=$(curl -sk -X POST https://vcenter.lab.local/api/session \ -u 'administrator@vsphere.local:<sso-pw>' | tr -d ' ') curl -sk -H "vmware-api-session-id: $SID" https://vcenter.lab.local/api/vcenter/datastore curl -sk -H "vmware-api-session-id: $SID" \ "https://vcenter.lab.local/api/vcenter/network?types=DISTRIBUTED_PORTGROUP"`

6.1 Confirm the symptom and read the authoritative error

1. Pre-flight: `GET /v1/resource-locks` must be **0** before triggering (else clear per the Resource-Lock DB-Cleanup runbook).
2. Trigger: `POST /v1/backups/tasks` body `{"elements":[{"resourceType":"SDDC_MANAGER"}]}` → note the task `id`.
3. Watch: `GET /v1/tasks/<id>` — see which sub-task FAILS.
4. Read the real error from that run's SOS log (newest dir): `D=$(ls -1dt /var/log/vmware/vcf/sddc-support/backups-* | head -1) grep -nE "KeyError|export_metadata|Exception|403|Unauthorized" "$D/sos.log" | tail -25 -403 / Unauthorized ON BackupSDDCManagerConfiguration → Phase A. - KeyError: 'primaryDatastoreName' ON BackupSDDCManagerSystemConfiguration → Phase B. - KeyError: 'name' ON BackupSDDCManagerSystemConfiguration → Phase C.`

6.2 Phase A — recreate the wiped NSX service account

Full detail is in the companion runbook "**Recreating a Missing SDDC Manager → NSX Service Account**" (v1.0). Summary of every step:

1. Read SDDC's stored NSX credential (on the appliance): `curl -k -u admin:<sddc-appliance-pw> https://127.0.0.1/v1/system/credentials/service` → find `svc-sddc-manager-nsx-vip-9971`; its `secret` field is the password.
2. Confirm it is missing on NSX (auth as NSX admin): `GET https://nsx-vip.lab.local/api/v1/node/users` (no such user), `/api/v1/aaa/local-users` (empty), `/api/v1/trust-management/principal-identities` (empty).
3. Create it: `POST https://nsx-vip.lab.local/api/v1/node/users?action=create_service_user` body `{"username":"svc-sddc-manager-nsx-vip-9971","password":"<the secret>"}` → returns a `userid` (e.g. `11000`), `password_reset_required:false`.
4. The create auto-adds an `auditor` role binding. `GET /api/v1/aaa/role-bindings` → find that user's binding `id` → `PUT /api/v1/aaa/role-bindings/<binding-id>` with `"roles":[{"role":"enterprise_admin"}]`.

5. Verify: `curl -k -u svc-sddc-manager-nsx-vip-9971:<secret> https://nsx-vip.lab.local/api/v1/cluster/status` → **HTTP 200**.
6. `systemctl restart operationsmanager`. The 403 clears and sub-task 3 passes.

Do NOT use Password Management → **Remediate** for a *missing* account — it fails "*Failed to get NSX user details*" and jams 4 stale locks. Remediate only works when the account still exists.

6.3 Phase B — correct the cluster's primary-datastore MoRef

1. Get the current vSAN datastore MoRef from vCenter (\$6.0) → e.g. `vcenter-cl01-ds-vsan01 = datastore-16`.
2. Get the SDDC cluster id + confirm the null name: `GET /v1/clusters` → `id=dc5f4a5b-...`, `primaryDatastoreName:null`, `primaryDatastoreType:VSAN`.
3. On the appliance as root, open the platform DB with the trust-auth workaround (PostgreSQL 15, `/data/pgdata`): `cp /data/pgdata/pg_hba.conf /data/pgdata/pg_hba.conf.bak sed -i -E 's/(scram-sha-256|md5)/trust/g' /data/pgdata/pg_hba.conf su - postgres -c "/usr/pgsql/15/bin/pg_ctl reload -D /data/pgdata" grep -E 'trust|scram-sha-256|md5' /data/pgdata/pg_hba.conf | grep -v '^#' Every host ... 127.0.0.1/32 and ::1/128 rule must now read trust before continuing.`
4. Confirm the stale value, then update the single row (expect `UPDATE 1`): `su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -c \"SELECT id,name,primary_datastore_name,primary_datastore_source_id,primary_datastore_type FROM cluster;\"" su - postgres -c "PAGER=cat psql -h 127.0.0.1 -d platform -c \"UPDATE cluster SET primary_datastore_source_id='datastore-16' WHERE id='dc5f4a5b-4ab2-41d9-9e7a-5e7d84ab2712';\""`
5. **Restore** `pg_hba.conf` immediately (never leave trust enabled): `cp /data/pgdata/pg_hba.conf.bak /data/pgdata/pg_hba.conf su - postgres -c "/usr/pgsql/15/bin/pg_ctl reload -D /data/pgdata" grep -nE 'scram-sha-256|md5|trust' /data/pgdata/pg_hba.conf | grep -vE '^[0-9]+:[[:space:]]*##' Every host rule must read scram-sha-256 again.`
6. `/opt/vmware/vcf/operationsmanager/scripts/cli/sddcmanager_restart_services.sh` (all services, ~2–5 min).
7. Verify: `GET /v1/clusters` → `primaryDatastoreName` now resolves. Re-running the backup clears the `primaryDatastoreName` `KeyError`.

6.4 Phase C — correct the VDS port-group MoRefs

1. Map transport type → current vCenter port-group MoRef (from `GET /api/vcenter/network?types=DISTRIBUTED_PORTGROUP`, \$6.0). On 2026-07-10: VM_MANAGEMENT → `dvportgroup-49` (pg-vm-mgmt), MANAGEMENT → `45` (pg-mgmt), VMOTION → `46` (pg-vmotion), VSAN → `47` (pg-vsan).
2. Find the VDS inventory id + the stale sourceids: `curl -s http://localhost/inventory/vds | python3 -m json.tool | head -60` On 2026-07-10: VDS id `72caf71e-ba7d-4ad4-8514-f42bb99c419b`; stale sourceids `dvportgroup-56-59`.
3. GET the record (a JSON array), strip to a single object, correct the 4 sourceids, PUT it back — **short lines only** (the console mangles long pastes): `cd /tmp u=http://localhost/inventory/vds i=72caf71e-ba7d-4ad4-8514-f42bb99c419b curl -s "$u?id=$i"|sed 's/^\[//;s/\]$//>v.json sed -i s/dvportgroup-59/dvportgroup-49/ v.json sed -i s/dvportgroup-56/dvportgroup-45/ v.json sed -i s/dvportgroup-57/dvportgroup-46/ v.json sed -i s/dvportgroup-58/dvportgroup-47/ v.json curl -s -XPUT -H content-type:application/json -d@v.json "$u/$i";echo curl -s "$u?id=$i"|grep -o '"sourceId": "[^"]*"'` - The GET returns a JSON **array**; the PUT body must be a single **object** (the `sed` bracket-strip) or it 400s "*Expected BEGIN_OBJECT but was BEGIN_ARRAY*". -

- The final `grep` must show `dvportgroup-49/47/46/45` — confirms the PUT persisted. - The port-group **name** will NOT appear in the inventory GET (derived, not stored) — do not chase it here; verify via the backup.
4. `/opt/vmware/vcf/operationsmanager/scripts/cli/sddcmanager_restart_services.sh`.

6.5 Verify end-to-end

1. Wait for warmup (~2–5 min; `POST /v1/tokens` authenticates again). `GET /v1/system/health` may still read **500** — that is the NSX-dependent health aggregation and does **not** block backups.
2. Confirm `GET /v1/resource-locks = 0`, then `POST /v1/backups/tasks`.
3. `GET /v1/tasks/<id>` until terminal — **all 8 sub-tasks SUCCESSFUL**, including `BackupSDDCManagerSystemConfiguration` and `SaveSDDCManagerBackup` (the SFTP write); locks return to 0.
4. The scheduled backup (same code path) now succeeds too. Take a fresh **no-memory** snapshot as a clean restore point.

6. Recommended Remediation

1. **Preferred — supported inventory reconcile:** run the async patch tool in sync mode, which re-reads live vCenter and repairs the stored MoRefs/attributes across the inventory: `./vcf-async-patch-tool --sync --sddc SSOUser administrator@vsphere.local --sddcSSHUser vcf` Requires the **entitled Broadcom download** (pending activation of the licensing channel).
2. **Alternative — Broadcom Support case**, per KB 429861 (the documented path for the `export_metadata` `Key Error`).
3. **After reconcile:** re-run `POST /v1/backups/tasks`, watch all 8 sub-tasks to `SaveSDDCManagerBackup` completing to the SFTP target, and confirm `GET /v1/system/health` returns 200 with a populated NSX VIP.

The two fixes already applied (§4) remain valid and reduce the reconcile's remaining work; they are not undone by the tool.

6.6 Provenance — what was documented vs. figured out

Diagnosis leaned on KBs; the working fixes were largely ours. Broadcom KBs named the error signatures and pointed the direction, but the actual repair — especially the port-group discovery and the *"names derive from MoRefs"* mechanism — was reverse-engineered here (disassembling the backup `.pyc`, probing the internal inventory API). KB 429861 (the exact backup `KeyError`) explicitly routes to Support with no published fix.

Element	Source
NSX 403 / missing-account diagnosis	Documented — KBs 314647, 420736, 327195
NSX account recreation (<code>create_service_user</code> + role PUT)	Figured out — no published procedure; assembled from the live NSX API spec
"Remediate jams 4 locks for a missing account"	Figured out (the hard way)
Datastore <code>KeyError</code> error signature	Documented — KB 429861 (its fix = "contact Support")

Element	Source
Datastore <code>primary_datastore_source_id</code> UPDATE	Semi — KB 444862 documents it for host-add, applied here
"Names derive from MoRefs" insight	Figured out — proven via the datastore fix
Port-group <code>KeyError</code> field (which object)	Figured out — disassembled the backup <code>.pyc</code>
Port-group MoRef fix via inventory PUT	Semi — KB 411099 documents the PUT pattern (edge clusters)
Array→object PUT-body requirement	Figured out (from the 400 error)

6.7 Known residual (not a backup blocker)

The `/v1/system/backup-restore/backups` and `/v1/system/health` endpoints still return **HTTP 500** — the operationsmanager health/backup-status *aggregation* is degraded by the **remaining** stale inventory refs (e.g. the vCenter `datastoreForVmDeploymentSourceId = datastore-25`, and the WatermarkService "Deployment Info Details" lookup). This does **not** affect the backup itself (proven complete), but it means status-reading tools (including the VCF Health Check app) cannot confirm backups via that endpoint — they will report "*backup endpoint not available (HTTP 500)*". Clearing it requires reconciling the residual MoRefs (async patch tool `--sync`, or the same manual method as §6.3/§6.4 applied to the `vcenter` record).

7. Verified Reference Facts

Item	Value
Cluster <code>vcenter-cl01</code> (SDDC id)	<code>dc5f4a5b-4ab2-41d9-9e7a-5e7d84ab2712</code> (vCenter MoRef <code>domain-cl0</code>)
vSAN datastore	<code>vcenter-cl01-ds-vsan01 = datastore-16</code> (was <code>datastore-25</code>)
VDS	<code>dvs-54</code> , inventory id <code>72caf71e-ba7d-4ad4-8514-f42bb99c419b</code>
NSX VIP / node	<code>nsx-vip.lab.local 192.168.1.70</code> / <code>nsx-manager.lab.local 192.168.1.71</code>
Backup module	<code>/opt/vmware/sddc-support/backup/sddcmanagerbackup.pyc</code> → <code>export_metadata</code>
Per-run SOS log	<code>/var/log/vmware/vcf/sddc-support/backup-<ts>-<pid>/sos.log</code>
Internal inventory API (localhost, no auth)	<code>curl http://localhost/inventory/{clusters,hosts,vcenters,vds}</code> (GET by id uses <code>?id=</code> ; PUT uses <code>/id</code> with an object body)
Platform DB	PostgreSQL 15, <code>/data/pgdata</code> ; trust-auth workaround per the Resource-Lock runbook
Relevant KBs	429861 (backup <code>KeyError</code> →Support), 444862 (<code>primary_datastore_source_id</code>), 327195 (credential store), 411099 (inventory PUT pattern), 313210 (recovery scripts)

8. Document Information

Title	SDDC Manager Backup Recovery — Stale Inventory MoRefs After an Out-of-Band Redeploy
-------	---

Author	Virtual Control LLC
Issued	July 10, 2026
Version	1.1
Classification	Internal
Platform	VMware Cloud Foundation 9.0.1 (SDDC Manager 9.0.1.0, PostgreSQL 15)
Related Documents	RUNBOOK — Recreating a Missing SDDC Manager → NSX Service Account (2026-07-10); RUNBOOK — SDDC Manager Resource-Lock & Post-Redeploy DB Cleanup (2026-06-09)
Distribution	Lab notebook

9. Revision History

Version	Date	Notes
1.0	2026-07-10	Initial issue. Captured the two-layer root cause (wiped NSX service account + systemic stale MoRefs), the derive-name-from-MoRef mechanism proven via the datastore fix and the <code>export_metadata</code> disassembly, the two verified fixes applied (NSX account recreate; <code>primary_datastore_source_id</code> 25→16), the remaining stale-MoRef inventory, and the decision to hand off to the supported <code>vcf-async-patch-tool --sync</code> reconcile rather than continue manual patching.
1.1	2026-07-10	RESOLVED. Corrected the 4 VDS port-group MoRefs (<code>dvportgroup-56-59</code> → <code>45-49</code> , inventory PUT) — the last blocker — and the backup then completed all 8 sub-tasks SUCCESSFUL to SFTP (entitled <code>async-patch-tool</code> not needed). Rewrote §5 as the resolution, added the full reproducible recovery procedure (§6, Phases A–C + verify, every command), the provenance table (§6.6, documented vs. figured out), and the known residual (§6.7 — the <code>/v1/system/backup-restore/backups</code> + <code>/v1/system/health</code> 500 from remaining stale refs, which makes status-readers like the Health Check app WARN even though the backup is good).